Facial Recognition Using Hybrid Technologies Based on Independence Testing

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Abstract. A facial recognition system is a technology capable of identifying or verifying a person from a digital image by comparing facial features with a database and has many applications such as home security and access control, finance, public safety and marketing, etc. Among these, the field of Home Security is particularly important for protecting family safety, preventing property loss, and ensuring psychological stability in daily life. Home security consists of a complex combination of various technologies, among which facial recognition technology requires high accuracy, a low false positive rate, and fast processing speeds as essential requirements. Furthermore, recent advancements in artificial intelligence have significantly improved the performance of facial recognition technology, establishing it as an indispensable core function. However, the reality is that applying face recognition technology in real-world environments is difficult because recognition results vary significantly depending on the diversity of facial images due to environmental changes, differences in facial structures between the East and West, and changes in facial expressions and gaze. Meanwhile, various methods to address these issues have recently been proposed. Representative examples include a hybrid approach combining computer vision and deep neural networks, and independence testing techniques that analyze statistical associations between visual variables, features, or data distributions of face images. In this paper, we proposed a methodology that organically links a hybrid approach combining computer vision and deep neural networks with independence testing and validated its effectiveness using the Labeled Faces in the Wild (LFW) dataset.

Keywords: Home Security, Facial Recognition, Hybrid Technology, Combination of Computer Vision and Deep Learning, Independence Testing

1 Introduction

Based on the advancement of artificial intelligence (AI) and deep learning, the latest facial recognition technology is maximizing accuracy and convenience through features such as remote detection, real-time processing, and contactless authentication. It is rapidly expanding into various industries, including security, financial digital identity verification, and smart access control.

Facial recognition technology used in home security identifies individuals using biometric(face) data. It captures photo images in real time to extract unique facial features into a digital facial fingerprint, which is then compared to a trusted database. The primary purpose is to automate access control, reduce false alarms, and immediately notify the homeowner in the presence of unfamiliar visitors or potential intruders.

In home security, facial recognition technology is being utilized as a key means to accurately identify residents and outsiders without the risk of losing keys or passwords, thereby blocking unauthorized intrusions in real time. It is producing significant ripple effects, such as improved ease of use, enhanced security accuracy, and increased connectivity with smart homes.

However, facial recognition technology in home security still has many factors that limit stable recognition results. These can be broadly divided into two categories: 1) bio-mechanical diversity of facial images and 2) physical changes due to environmental changes.

Examples of the first factor include structural differences in facial features between Eastern and Western people, diversity in facial expressions and behaviors, and diversity in style and adornment.

Furthermore, examples of the second factor include changes in lighting and light sources, changes in camera angles and poses, and changes in background and context.

Various technologies have been developed to solve these problems from the early stages of research to the present, the representative of which is Hybrid technology that combines traditional Computer Vision technology with Deep Learning technology, an application of Neural Network technology, and independence testing techniques that analyze statistical associations between visual variables, features, or data distributions of face images.

The purpose of this study is to develop stable and robust facial recognition technology by resolving the aforementioned problems in face recognition, and to this end, we proposed a methodology that organically links a hybrid approach combining computer vision and deep neural networks with independence testing and validated its effectiveness using the Labeled Faces in the Wild (LFW) dataset.

2 Related Work

Classical recognition. Eigenfaces [1] projects faces onto PCA components; Fisherfaces [2] adds LDA to separate classes; LBPH [3] compares local binary-pattern histograms per region. All three ship in OpenCV [10], need no GPU, and produce models from kilobytes to a few megabytes.

Deep recognition. FaceNet [6] mapped faces to an embedding space with triplet loss. SFace [4] uses a sigmoid-constrained hypersphere loss and is small enough to run on edge hardware, so we use it as the hybrid's DL engine.

Detection. Viola-Jones Haar cascades [5] remain the classical baseline; YuNet [7] is a millisecond-scale CNN detector that also returns the five landmarks DL recognizers use for alignment.

Evaluation. Biometric practice separates TAR, FAR, and FRR [9]. LFW [8] tests verification on fixed pair lists; independence testing instead yields the whole impostor distribution, so a threshold ties to an exact error count. For claims that two recognizers complement each other we adopt the standard yardsticks rather than invent our own: pairwise diversity measures (Yule's Q, double-fault [11]) and McNemar's paired test [12].

3 Method

3.1 System overview

LS-Face processes one camera frame as follows. A shared YuNet front-end returns one face box, a confidence, and five landmarks. The frame first takes the cheap path: LBPH (grayscale 100×100 crop, Tan-Triggs illumination normalization) predicts an identity and a distance d. A gate then decides whether that answer can be trusted. If yes, LBPH's decision stands and the DL model never runs. If not, the frame escalates: SFace aligns the face to 112×112 with the landmarks, extracts a 128-D (512-byte) embedding, and matches it against per-identity mean embeddings by cosine similarity. A no-accelerator fallback (LBPH alone) engages automatically if the DL gallery is absent.

3.2 Offline Candidate Screening and Component Selection

Component selection is an offline development stage, distinct from the frame-level escalation gate. Classical candidates are selected on La Salle DB1 only: a deterministic subset of its training images fits each model, a disjoint La Salle calibration subset sets that model's native-score operating point, and the held-out La Salle test split and its 41-modification probes remain untouched until evaluation. We considered the classical recognizers LBPH, Eigenfaces, and Fisherfaces, and the learned embedding models SFace, FaceNet, and ArcFace. Since their score scales are not comparable, raw distances were never compared across families. Instead, each candidate was assessed on its native score using cross-identity impostor comparisons, held-out genuine recognition, robustness under the 41 image modifications, runtime, and per-identity representation size. This offline selection protocol is separate from later LFW-based calibration that freezes deployed LBPH--SFace cascade thresholds.

For the classical pool, a scripted constrained selector requires acceptable true-acceptance performance at a fixed false-acceptance operating point and sufficient runtime, then ranks eligible models by modification robustness, with true-acceptance rate and model size as tie-breakers. Its strict recorded rule has no candidate that meets every project constraint: LBPH supplies the needed verification and robustness but its 64-kB histogram exceeds the nominal 1-kB representation budget, while Eigenfaces and Fisherfaces fail the genuine-acceptance requirement. LBPH is therefore described as the selected fast-path recognizer, not as the winner of an unconstrained six-model optimization; the deep gallery supplies the compact enrollment representation.

The learned-model screening used a common YuNet alignment and brightness-normalization front end. The available artifacts establish a 128-dimensional, 512-byte SFace embedding, compared with 512-dimensional, 2,048-byte FaceNet and ArcFace representations. SFace was retained as the deployment-compatible escalation model. This is a resource-constrained engineering choice, not a demonstrated accuracy win over FaceNet and ArcFace: archived learned-model results use model-specific thresholds, and their reported true-positive values allow self-matching. After component selection, subsequent cascade experiments evaluate the frozen LBPH--SFace configuration rather than retuning it on reported test outcomes.

3.3 Independence testing and threshold rule

Take N identities with one image each. Comparing every unique pair gives

$C = N(N - 1) / 2$ unique comparisons (1)

all impostor pairs by construction. Sort the C distances ascending. To operate at a target rate FAR*, choose the rank

$$k = \lceil \mathrm{FAR}^* \cdot C \rceil \qquad (1)$$

and set θ to the k-th smallest impostor distance. The rank sets the target false-accept count; after thresholding, we recompute the realized FAR from all pairs that satisfy the deployed comparison rule, including ties. The deployed LBPH acceptance threshold is derived on LFW DB1: N=5,749 gives C=16,522,626 unique impostor pairs; rank 165 produces 9.986 ppm FAR. A La Salle DB1 sweep (N=28) is too small to resolve a 10-ppm operating point. The full formalism is in docs/archive/report_docs/independence_test/MATHEMATICAL_FOUNDATION.md in the repository.

3.4 The escalation gate

Let d_1 and d_2 be the best and second-best LBPH distances. Let τ_a be the frozen LBPH acceptance threshold and $\tau\_r > \tau\_a$ the frozen escalation/reject threshold.

(i) a quality flag fires (blur, low light, sensor noise, off-pose, or too-small face, measured on the crop LBPH already holds); (ii) the score is ambiguous: $\tau\_a < d_1 < \tau\_r$; (iii) the top-two margin is thin:

$$(d_2 - d_1)/d_1 < m\_min \qquad (2)$$

The margin is relative because LBPH training distances are near zero by memorization; an absolute margin fitted on training data escalates every held-out frame. A quality flag deliberately overrides a confident LBPH score: the corrupted regimes are exactly where LBPH confidence is least trustworthy. If nothing fires, $d_1 \leq \tau\_a$ accepts on LBPH and $d_1 \geq \tau\_r$ rejects.

3.5 Robustness: the 41-modification accuracy ratio

Every original image receives 41 deterministic (modification, level) variants across 12 types: brightness up/down, contrast up/down, gamma up/down, Gaussian noise, Gaussian blur, motion blur, rotation, zoom, occlusion. A modified probe matches when the recognizer outputs the correct identity within the deployed threshold. With M probes and K matches,

$$AR = K / M \quad (3)$$

averaged per modification over its levels, then over modifications. The suite is seeded per (image, modification, level), so CV, DL, and hybrid score bit-identical probes.

Testing complementarity directly

Complementarity cannot be inferred from aggregate accuracy alone. We test a three-link argument on the same held-out probes: potential fallback value, routability, and measured utility. First, paired thresholded decisions are counted as both right, LBPH only right, SFace only right, or both wrong. The conditional rescue rate is the share of LBPH failures corrected by SFace. Since 41 deterministic transforms repeat each source image, these row counts are descriptive conditions, not independent trials; no row-level significance claim is made.

Second, the cascade is useful only if LBPH exposes a signal before SFace runs. We therefore score the best distance and negative relative margin against threshold-free LBPH Rank-1 error and report ROC AUC, where 0.5 denotes blind routing. The deployed rule is evaluated separately against thresholded LBPH system failure, with its recall, false-routing rate, and precision reported only for probes that yield a gate signal.

Third, routing must justify its cost. The frozen gate is compared with always-LBPH and always-SFace anchors on thresholded correct-identity acceptance, escalation rate, and recognition-stage time from the same records. A selective cascade should retain the learned tier's benefit while avoiding enough learned-tier calls to offset the LBPH overhead. If it matches SFace but runs slower, the evidence supports fallback behavior rather than an efficiency advantage.

Frozen cascade configuration across databases

Separately tuned values across four databases would show tunability, not transfer. We freeze one configuration before cross-database evaluation: $\tau\_a = 67.03325520645528$ from LFW DB1's unique-pair LBPH sweep (rank 165; 9.986 ppm); $\tau\_r = 140.13$ from the LFW escalation trade-off; $m\_min = 0.05$ is policy-set; and SFace uses its deployed L2 rule, 1.0313. The harness records the threshold-file SHA-256, and the same configuration is applied unchanged across database legs. Enrollment is always clean originals; only probes are modified. Each database answers one question (Table 1):

Table 1. Evidence matrix

Database	Test	What it proves

La Salle DB1 (28 ids, clean)	independence sweep, 10 seeded repeats	in-domain FAR at the frozen thresholds
La Salle DB2 (41 mods)	accuracy ratio, CV / DL / cascade	robustness under degradation; per-modification winners
LFW DB1 (5,749 ids, clean)	independence sweep	out-of-domain transfer with real statistical power
LFW DB2 (41 mods, 1 image/id)	independence sweep on modified probes	degradation and identity separation jointly, out of domain

All identification and verification numbers on LS-DB1/LS-DB2 use closed-set enrollment: gallery and probes share the same 28 identities and are image-disjoint (10 gallery / 2 probe images per identity), mirroring a gate that admits only enrolled subjects. Robustness to unseen identities is measured as impostor rejection against the LFW legs; open-set identification is out of scope.

4 Experiments and Results

4.1 Algorithm Selection on La Salle DB1

Classical candidates were fitted on 224 images, calibrated on 56 disjoint images, and evaluated once on 56 untouched LSDB test images. The calibration set supplied 1,512 cross-identity scores; the 15th error score yielded realized FAR 0.992%. LBPH achieved the highest test TAR (96.43%) and Rank-1 accuracy (100.00%) and was selected as the classical fast path. Its 64 KB representation remains an explicit engineering exception.

Table 2. LSDB classical candidate selection. TAR is thresholded at each model's calibration-only LSDB operating point.

Recognizer	**TAR (%)**	**Rank-1 (%)**	**Feature size**
LBPH	96.43%	100.00%	64 KB
Eigenfaces	41.07%	75.00%	896 B
Fisherfaces	51.79%	58.93%	108 B

A fresh DL-only campaign replaced the legacy self-match test. SFace, ArcFace, and FaceNet used the deterministic 224/56/56 LSDB cohort; each model's rank-15 acceptance edge came only from 1,512 calibration cross-identity scores (realized FAR 0.992%). Native score scales remained model-specific and were not mixed with the classical ranking.

Table 3. DL-only candidate selection on LSDB. TAR uses each model's rank-15 calibration edge (0.992% realized FAR).

Candidate	TAR (%)	Rank-1 (%)	Feature
SFace	100.00%	100.00%	512 B
ArcFace	96.43%	100.00%	2,048 B
FaceNet	100.00%	100.00%	2,048 B

SFace and FaceNet tied at 100.00% held-out TAR and Rank-1; SFace won the recorded feature-size tie-break (512 B versus 2,048 B). ArcFace retained 100.00% Rank-1 but reached 96.43% TAR. Thus, SFace is the learned tier, while LBPH remains the separately selected classical fast path.

4.2 Independence Test: Frozen Operating Points

LFW DB1 fixes the deployed native-score operating points before LSDB evaluation: LBPH tau_accept = 67.0333 (rank 165 of 16,522,626 unique impostor pairs; 9.986 ppm), SFace L2 = 1.0313 with cosine >= 0.363, and tau_reject = 140.13.

Table 4. Final frozen operating points.

Boundary	Source / native scale	Role
LBPH tau_accept	LFW; predict_collect	accept
SFace L2	LFW; SFace L2	escalated accept
LBPH tau_reject	LFW trade-off; predict_collect	permissive reject edge

Note. Frozen LFW-derived values: LBPH tau_accept = 67.03325520645528; SFace L2 = 1.0313 with cosine >= 0.363; and LBPH tau_reject = 140.13. They are fixed before LSDB evaluation and not re-tuned on LSDB.

The frozen-boundary plot in Fig. 1 shows the frozen boundaries against the fresh LSDB clean-impostor distribution; it is not a re-calibration.

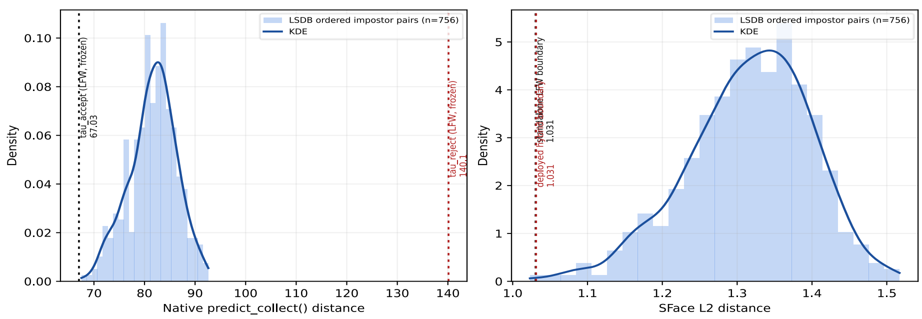

Fig. 1. Histograms and KDE curves for fresh LSDB clean impostor pairs. Long-dotted lines mark frozen LFW operating points.

4.3 LFW2 Robustness Evaluation

Table 5 summarizes gallery/probe-disjoint LFW2 1-to-N identification robustness at the frozen deployment thresholds. Across all 41 modifications, SFace and the cascade retain 80.65% AR, whereas LBPH alone retains 1.41%. The cascade escalates 97.51% of probes to SFace; isolated latency is reported from a 575-identity subset.

Table 5. LFW2 1-to-N identification robustness at frozen deployment thresholds.

Mode	AR (%)	Latency (ms)	FAR (%)	Escalation (%)
Classical CV (LBPH)	1.41	72.49	0.0010	N/A
Deep Learning (SFace)	80.65	84.36	~0.0010	N/A
Hybrid Cascade	80.65	82.54	≤0.0020	97.51

Overall AR and escalation are means across 41 modifications (5,749 enrolled identities; 1,680 probes; strict no-face policy). Latencies are isolated means from a 575-identity/172-probe subset. Standalone FAR values are LFW1 9.986-ppm calibrations; SFace is approximate at the deployed boundary. Cascade ≤0.0020% is a conservative union bound, not a measured joint FAR.

Complementarity on Held-Out LSDB-DL41 Probes

The evaluation follows the three-link argument above. It uses 56 image-disjoint held-out LSDB source probes, two for each of 28 enrolled identities, and 41 deterministic transformations per source, yielding 2,296 correlated probe conditions. The LFW-derived thresholds and gate were frozen before scoring, and detector failures were handled strictly. Table 6 therefore pairs each number with the exact conclusion it can support.

Table 6. Logic-to-data audit of SFace rescue and gate routing on held-out LSDB-DL41 probes.

Question	Recorded data	Supported reading
Fallback value	Both right 707; LBPH only 0; SFace only 1,296; both wrong 293	One-way rescue; not mutual
Rescue magnitude	1,296 / 1,589 LBPH errors = 81.56%	Descriptive correlated transforms
Can the gate rank risk?	Distance AUC 0.95019; negative-margin AUC 0.95319; 2,060 scored, 444 Rank-1 errors; 236 no signal	Strong within-suite discrimination
Does routing catch scored failures?	1,353 / 1,353 failures routed; 289 / 707 correct cases also routed; precision 82.40%	Full failure recall; substantial over-routing
Accuracy-cost result	Deployed: 87.24% AR, 71.52% escalated, 11.96 ms; candidate: 87.24%, 59.23%, 10.81 ms; SFace: 87.24%, 8.33 ms	Less redundant routing; direct SFace remains faster

First, the value of the fallback is asymmetric. SFace corrected 1,296 of the 1,589 thresholded LBPH failures (81.56%), whereas LBPH supplied no thresholded success that SFace lacked. The 293 both-wrong conditions identify the remaining failure region for these two final decisions; they are not a universal ceiling for every possible score-level fusion. Because transformations repeat source images, the paired counts are reported descriptively rather than as 2,296 independent observations.

Second, the gate exposes a useful risk signal, but its accept-side quality overrides cause redundant routing on this battery. Among 2,060 probes with gate signals, distance and negative margin separated 444 threshold-free LBPH Rank-1 errors with AUCs of 0.95019 and 0.95319. The deployed rule routed all 1,353 scored LBPH failures and 289 of 707 scored LBPH-correct cases. A post-hoc accept-protection replay kept the deployed low-margin trigger and every rule above tau_accept unchanged while treating accept-side quality flags as telemetry; it reduced those correct-case escalations to 7 of 707. Because the same transformed probes motivated and evaluated the candidate, this is a canonical descriptive ablation rather than independent policy validation.

Third, accept protection improves the deployed route without establishing a speed advantage over SFace. The replay preserved 87.24% thresholded correct-identity acceptance while reducing escalation from 71.52% to 59.23% and the stored recognition-stage arithmetic mean from 11.96 to 10.81 ms. Direct SFace still reached the same acceptance at 8.33 ms, and LBPH-only reached 30.79% at 5.25 ms. These single-pass stored-stage timings exclude detection and I/O; the result supports simpler routing, not a Pareto improvement or a runtime gate change.

Discussion

The experiments support a division of labor with a clear negative boundary. LSDB selection retains LBPH as the tested classical fast path and SFace as the compact learned tier; the held-out DL41 stress test shows one-way SFace rescue and gate AUCs near 0.95. The descriptive accept-protection replay then isolates the main source of redundant routing: preventing quality flags from overriding an LBPH decision already inside tau_accept preserves 87.24% AR while reducing escalation to 59.23%. However, the resulting 10.81 ms arithmetic mean remains slower than direct SFace. The contribution is therefore interpretable fallback and a simpler candidate rule, not mutual accuracy gain or a Pareto-efficient cascade.

The scope also limits generalization. The LSDB-DL41 rows are correlated transformations of 56 known-genuine source probes, 236 no-signal cases are outside the scored gate rates, and no unknown-query FPIR can be measured. The stored timing is recognition-stage arithmetic rather than end-to-end latency, while LFW2 drives 97.51% escalation at the frozen gate. The runtime gate remains unchanged. Open-set identification and target-device testing, including Raspberry Pi 5 measurements, are explicitly outside the present paper's scope.

Conclusion

This paper presented LS-Face, a gated LBPH-to-SFace recognizer whose component selection, threshold calibration, and held-out evaluation remain separate evidence stages. On LSDB-DL41, SFace recovered 81.56% of thresholded LBPH failures and gate signals ranked LBPH Rank-1 risk with AUCs near 0.95. A canonical descriptive accept-protection replay preserved 87.24% AR while reducing escalation from 71.52% to 59.23% and the stored arithmetic mean from 11.96 to 10.81 ms; direct SFace remained faster. The runtime gate is unchanged, and the result is a same-data routing observation rather than independently validated deployment evidence. Open-set and target-device measurements are outside scope.

5 Acknowledgments

The authors thank the external deep-learning team for providing the candidate-model artifacts used in the offline screening stage.

6 Disclosure of Interests

The authors have no competing interests to declare.

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